A nonsemisimple Specht module of weight one

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Scope of the counterexample. The printed definition says that Specht modules are semisimple when the p-restricted weight is at most $p/2$, and the conjecture asserts that the first counterexample, if any, has $n \geq 16$. We use the standard reading of a p-restricted partition and its block p-weight. For

$$p = 3, \qquad n = 3, \qquad \lambda = (2,1), \qquad F = \mathbb{F}_3,$$

the partition is 3-restricted, has empty 3-core and 3-weight one, but the actual Specht module $S_F^{(2,1)}$ is not semisimple. Thus it contradicts the printed semisimplicity assertion, already with $n < 16$. The customary James conjecture concerns decomposition and adjustment matrices, not this semisimplicity assertion; see [3]. This submission does not claim a counterexample to that customary conjecture. It also makes no claim against the $p \geq 2n$ range: here $3 < 2 \cdot 3$.

The partition, core and weight are computed. The native Young diagram has cells $(0,0), (0,1), (1,0)$. Its row lengths are $2, 1, 0, \ldots$, so each consecutive difference is less than 3. The whole diagram is a connected border strip of three cells, contains no 2×2 square, and can be removed to leave the empty diagram. The empty diagram has no further positive rim-hook removal. Moreover every k-step sequence of rim 3-hook removals ending at a diagram μ satisfies

$$|\mu| + 3k = 3. \tag{1}$$

A terminal sequence cannot have $k = 0$, since the original strip is removable. Hence every terminal sequence has $k = 1$ and $\mu = \varnothing$. This proves that the true 3-core is empty. The formal weight is the maximum length of genuine removal chains, with the bound supplied by the diagram's cardinality; (1) proves that this maximum is exactly one. In particular $2w_3(\lambda) = 2 \leq 3$. Neither core nor weight is assigned its desired value. These are the usual rim-hook and restricted-partition conventions in [1], printed p. 5.

Actual tabloids and polytabloids. Fix the positions of a tableau as $\begin{smallmatrix} 0 & 1 \\ 2 \end{smallmatrix}$. All tableaux are all bijective labelings of these three cells. Two tableaux are row equivalent precisely when they have the same bottom entry. The proof forms the genuine row-equivalence quotient and its bijection with $\{0,1,2\}$, and verifies the induced S_3 action. Thus the Young permutation module is $M = F^3$ with basis e_0, e_1, e_2 indexed by the bottom entry.

For a tableau t, its column stabilizer permutes the two labels $t(0), t(2)$ and fixes $t(1)$. The actual signed column sum is therefore

$$e_t = \sum_{g \in C_t} \operatorname{sgn}(g)\, e_{(gt)(2)} = e_{t(2)} - e_{t(0)}. \tag{2}$$

The formal construction defines C_t as a subgroup and proves that it preserves every actual column-label set. It uses the native permutation sign and proves (2) for all tableaux. This is the standard polytabloid construction over a ring from [2], printed p. 1.

The genuine Specht module is the augmentation module. By definition $S_F^{(2,1)}$ is the span of all polytabloids (2). Their coordinates sum to zero. Conversely, if $x_0 + x_1 + x_2 = 0$, then

$$(x_0, x_1, x_2) = x_1(e_1 - e_0) + x_2(e_2 - e_0),$$

and both displayed differences are actual polytabloids. Hence

$$S_F^{(2,1)} = V := \{x \in F^3 : x_0 + x_1 + x_2 = 0\}. \tag{3}$$

The proof establishes this equality and constructs the induced linear equivalence, with its equivariance under every permutation. The action is $(\rho(g)x)_i = x_{g^{-1}(i)}$.

A proper nonzero invariant line with no invariant complement. In characteristic 3, the line

$$C = F(1, 1, 1) \subset V$$

is invariant and nonzero. It is proper: $e_2 - e_0$ lies in V but is not constant. Let $c = (012)$, put $R = \rho(c)$ and $N = R - I$. For $x \in V$,

$$Nx = (x_2 - x_0,\ x_0 - x_1,\ x_1 - x_2).$$

Its three entries are equal, using $x_0 + x_1 + x_2 = 0$ and $3 = 0$. Thus $N(V) \subseteq C$. Also $Rx = x$ forces all three coordinates of x to be equal, so $\ker N = C$.

Suppose an S_3-invariant subspace W complemented C. For $w \in W$, $Nw \in W$ by invariance and $Nw \in C$ by the preceding calculation. Since $C \cap W = 0$, we have $Nw = 0$. Therefore $W \subseteq \ker N = C$, whence $W = 0$. This contradicts $C + W = V$ because C is proper. There is no invariant complement to C, so V, and hence the actual Specht module (3), is *not semisimple as an $F[S_3]$-module*.

Formal coverage and conclusions. The Lean development uses native Young diagrams, their actual finite cell sets, row-equivalence quotient tabloids, permutation sign, the full signed column sum, polytabloid span and an explicit S_3-equivariant augmentation equivalence. The invariant-line obstruction is stated with `Representation.IsSemisimpleRepresentation` and transferred to the native `IsSemisimpleModule` for the group algebra. Semisimplicity of the underlying F-vector space is not the assertion used. The separate rim-hook development proves core uniqueness and the real maximum chain count, then supplies the printed size and weight inequalities.

Together, these calculations give a counterexample under the stated standard block-weight interpretation at $(p, n) = (3, 3)$. They refute the source's printed definition-based assertion and its proposed lower size for a first such counterexample. They neither classify all Specht modules nor modify the usual meaning or status of the decomposition-matrix James conjecture.

The package pins Lean 4.33.0 and Mathlib `db584cd6d46c92f209a44c0f1c829460d327499d`. Its evidence records the source-bound build, all-namespace ordinary `#print axioms` audit and official fresh-kernel replay. No admitted proofs, extra axioms or `native_decide` are used. Production timings are not controlled efficiency measurements.

References

[1] Y. Jiang, *On some trivial source Specht modules*, arXiv:1812.07882, v2 (2021), printed pp. 5–6 (core, weight, restricted partitions and Specht modules).

[2] M. Fayers, *On the structure of Specht modules*, J. London Math. Soc. **67** (2003), 85–102. doi:10.1112/S0024610702003824, printed p. 1 of the author's manuscript (tabloids and column antisymmetrizer).

[3] M. Fayers, *James's Conjecture holds for weight four blocks of Iwahori–Hecke algebras*, J. Algebra **317** (2007), 593–633. Author's manuscript, printed p. 1 (decomposition/adjustment-matrix scope).